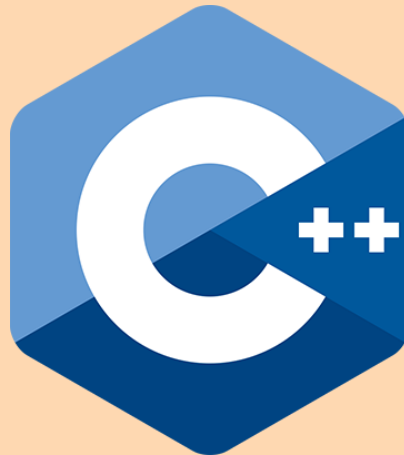


1. VARIABLES, TYPES, I/O

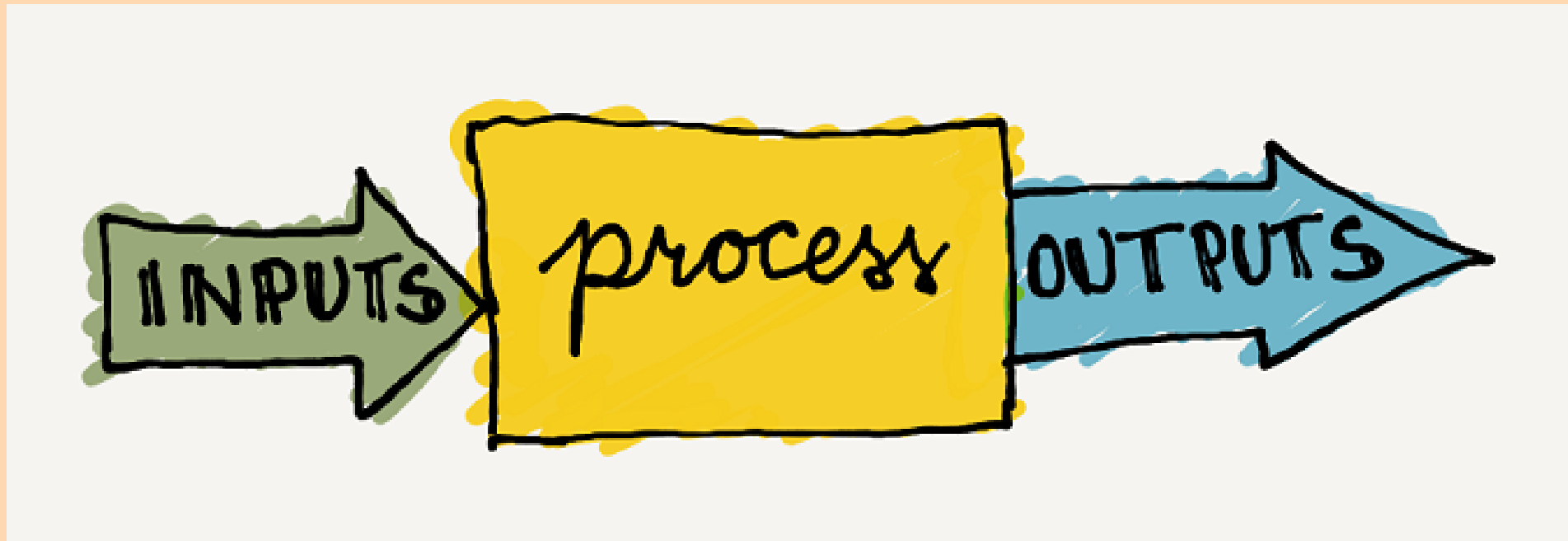
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Overview

Program = **Input** + **Data** + **Operations** + **Output**



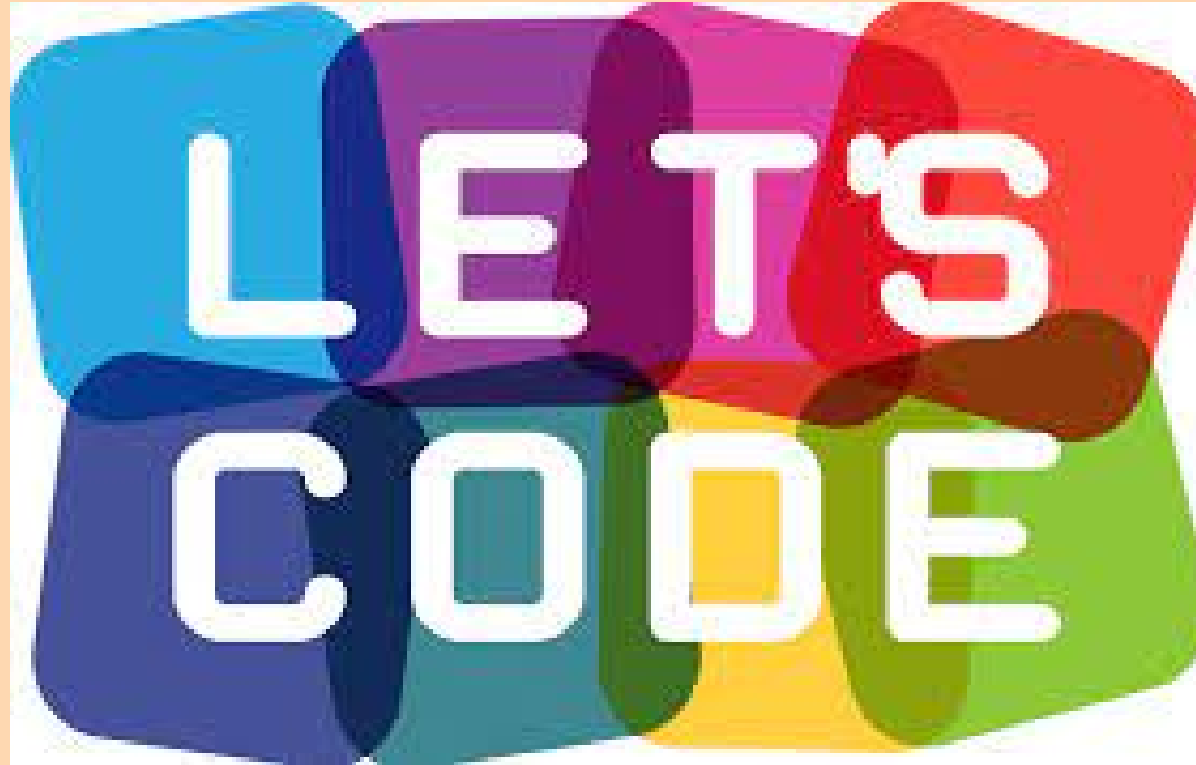
Overview

Variables (变量) Data Types (数据类型)

Operators (运算符) Expressions (表达式)

Assignment Statements (赋值语句)

I/O (Input/Output; 输入输出)



circle.c sphere.c

admin.c admin-cin.c

Circle

Given a **radius** (say 10) of a circle,
to compute its **circumference** and **area**.

$$L = 2\pi r \quad S = \pi r^2$$

- 每个结果各占一行
- 小数点后保留两位

Declaration/Definition (声明/定义)

```
int radius{10};
```

- Declare/Define a *variable* called `radius`.
- `radius` refers to a *location* in memory.
- The *type* of `radius` is `int` (integer).
- `radius` is *initialized* to 10.
- You can *assign* other integers to `radius`.

Identifiers (标识符)

```
int radius{10};
```

`radius` is an *identifier*.

Warning: Do *not* start with `_`, which are reserved by C++.

Always use `meaningful` identifiers in a `uniform` style!!!



```
int radius;
```

```
cout << radius;
```

Undefined Behavior (UB).

Operators & Expressions

```
double circumference{2 * PI * radius};
```

Sphere

Given a **radius** (say 100) of a sphere,
to compute its **surface area** and **volume**.

$$A = 4\pi r^2 \quad V = \frac{4}{3}\pi r^3$$

- 每个结果占 1 行
- 小数点后保留 4 位
- 每个结果至少占 15 字符, 左对齐
 - `_____ : surface_area`
 - `_____ : volume`

Input/output manipulators (操纵符)

```
#include <iomanip>
```

<https://en.cppreference.com/cpp/io/manip>

std::format

<https://en.cppreference.com/cpp/utility/format/spec>

Format Specification

```
{:[align][width][.precision][type]}
```

- **%d**: decimal `int`
- **%f**: `double`
- **%S**: `string` (not necessarily)
- **%C**: `char` (not necessarily)

```
{:[align][width][.precision][type]}
```

- <: left-justified
- >: right-justified
- ^: centered

```
{:[align][width][.precision][type]}
```

- minimum field width
- padded with spaces if it has fewer characters

```
{:[align][width][.precision][type]}
```

- %f: number of digits after .
- %s: maximum number of characters

A (Naive) Administration System

- Name (EN)
- Gender (F/M)
- Birthday (mm-dd-yyyy)
- Weekday (Xyz.)
- C++
- Music
- Medicine
- Mean (.d)
- Standard Deviation (.dd)
- Ranking (%)



For 罗大佑 only:

- 每组信息占一行
- 各项信息使用 `\t` 间隔
- 各项信息遵循特定格式要求

char

	_0	_1	_2	_3	_4	_5	_6	_7	_8	_9	_A	_B	_C	_D	_E	_F
0_ 0	NUL 0000	SOH 0001	STX 0002	ETX 0003	EOT 0004	ENQ 0005	ACK 0006	BEL 0007	BS 0008	HT 0009	LF 000A	VT 000B	FF 000C	CR 000D	SO 000E	SI 000F
1_ 16	DLE 0010	DC1 0011	DC2 0012	DC3 0013	DC4 0014	NAK 0015	SYN 0016	ETB 0017	CAN 0018	EM 0019	SUB 001A	ESC 001B	FS 001C	GS 001D	RS 001E	US 001F
2_ 32	SP 0020	! 0021	" 0022	# 0023	\$ 0024	% 0025	& 0026	' 0027	(0028	) 0029	* 002A	+ 002B	, 002C	- 002D	. 002E	/ 002F
3_ 48	0 0030	1 0031	2 0032	3 0033	4 0034	5 0035	6 0036	7 0037	8 0038	9 0039	: 003A	; 003B	< 003C	= 003D	> 003E	? 003F
4_ 64	@ 0040	A 0041	B 0042	C 0043	D 0044	E 0045	F 0046	G 0047	H 0048	I 0049	J 004A	K 004B	L 004C	M 004D	N 004E	O 004F
5_ 80	P 0050	Q 0051	R 0052	S 0053	T 0054	U 0055	V 0056	W 0057	X 0058	Y 0059	Z 005A	[005B	\ 005C	] 005D	^ 005E	_ 005F
6_ 96	` 0060	a 0061	b 0062	c 0063	d 0064	e 0065	f 0066	g 0067	h 0068	i 0069	j 006A	k 006B	l 006C	m 006D	n 006E	o 006F
7_ 112	p 0070	q 0071	r 0072	s 0073	t 0074	u 0075	v 0076	w 0077	x 0078	y 0079	z 007A	{ 007B	 007C	} 007D	~ 007E	DEL 007F

A `char` is actually an `int`.

C++ String

```
std::string first_name{"Tayu"};
```



未来的主人翁

Review

Program = **Input** + **Data** + **Operations** + **Output**

